

A simplified risk score for predicting postoperative nausea and vomiting: conclusions from cross-validations between two centers.



BACKGROUND: Recently, two centers have independently developed a risk score for predicting postoperative nausea and vomiting (PONV). This study investigated (1) whether risk scores are valid across centers and (2) whether risk scores based on logistic regression coefficients can be simplified without loss of discriminating power. METHODS: Adult patients from two centers (Oulu, Finland: n = 520, and Wuerzburg, Germany: n = 2202) received inhalational anesthesia (without antiemetic prophylaxis) for various types of surgery. PONV was defined as nausea or vomiting within 24 h of surgery. Risk scores to estimate the probability of PONV were obtained by fitting logistic regression models. Simplified risk scores were constructed based on the number of risk factors that were found significant in the logistic regression analyses. Original and simplified scores were cross-validated. A combined data set was created to estimate a potential center effect and to construct a final risk score. The discriminating power of each score was assessed using the area under the receiver operating characteristic curves. RESULTS: Risk scores derived from one center were able to predict PONV from the other center (area under the curve = 0.65-0.75). Simplification did not essentially weaken the discriminating power (area under the curve = 0.63-0.73). No center effect could be detected in a combined data set (odds ratio = 1.06, 95% confidence interval = 0.71-1.59). The final score consisted of four predictors: female gender, history of motion sickness (MS) or PONV, nonsmoking, and the use of postoperative opioids. If none, one, two, three, or four of these risk factors were present, the incidences of PONV were 10%, 21%, 39%, 61% and 79%. CONCLUSIONS: The risk scores derived from one center proved valid in the other and could be simplified without significant loss of discriminating power. Therefore, it appears that this risk score has broad applicability in predicting PONV in adult patients undergoing inhalational anesthesia for various types of surgery. For patients with at least two out of these four identified predictors a prophylactic antiemetic strategy should be considered.